Emergent Player Styles in AI-Simulated TRPG: From Innovation Instances to Ratified Design Patterns

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Abstract

Describing player behavior in tabletop role-playing games (TRPGs) systematically is difficult: sessions are unrecorded, behavior is attributed to player personality rather than character-sheet design, and no formal mechanism exists for promoting session surprises into reusable design patterns. We describe a pipeline that addresses this gap, applied across 49 AI-simulated Dungeons & Dragons 5e sessions spanning 7 complete campaigns. The pipeline promotes behavioral surprises to ratified *player styles* when 3 or more instances cluster thematically across 2 or more sessions. Seven styles were promoted: *sheet-deep-reader* (˜50 instances), *craft-witness* (˜45 instances), *npc-arc-completion trigger* (˜40 instances), *act-without-announcement* (˜25 instances), *verbal-documentation-arc* (˜15 instances), *documentary-witness* (˜12 instances), and *act-without-announcement intra-session* (˜8 instances). Cross-campaign validation against 12 research hypotheses (RV1–RV12) confirms that all 7 styles fire across all 7 campaigns and are not party-archetype-specific. Four PC-authored finale deliverables across 4 campaigns demonstrate that player styles accumulate toward campaign-completing acts when character-sheet design enables it.

Contents

1 Introduction **3**
1.1 The Contribution . . . 3
1.2 Why AI Simulation Is a Useful Laboratory . . . 3
1.3 The Player-Style vs. Player-Type Distinction . . . 4
1.4 Roadmap . . . 4

2 Related Work **4**
2.1 Player Typologies . . . 4
2.2 Emergent Narrative and Emergent Behavior . . . 5
2.3 AI Player Behavior in Games . . . 5
2.4 Design Patterns in Games . . . 5

3 Methodology **5**
3.1 The Promotion Pipeline . . . 5
3.2 Threshold Calibration . . . 6
3.3 The Seven Promoted Styles . . . 7
3.3.1 Sheet-Deep Reader (∼50 instances) . . . 7
3.3.2 Craft-Witness (∼45 instances) . . . 8
3.3.3 NPC-Arc-Completion Trigger (∼40 instances) . . . 8
3.3.4 Act-Without-Announcement, Cross-Session (∼25 instances) . . . 8
3.3.5 Verbal Documentation-Arc (∼15 instances) . . . 9
3.3.6 Documentary-Witness (∼12 instances) . . . 9
3.3.7 Act-Without-Announcement, Intra-Session (v1.12, ∼8 instances) . . . 9
3.4 PC-Authored Finale Deliverables . . . 10

4 Cross-Campaign Validation **10**
4.1 Research Hypotheses . . . 10
4.2 Results: Group 1 (Campaign 5, Professional Triggers) . . . 11
4.3 Results: Group 2 (Campaign 6, Duty Triggers) . . . 12
4.4 Results: Group 3 (Campaign 7, Situational-to-Relationship Shift) . . . 12
4.5 Aggregate Instance Counts . . . 13
4.6 Research Hypothesis Summary . . . 13

5 Discussion **13**
5.1 Styles Are Not Party-Archetype-Specific . . . 13
5.2 The Promotion Threshold Is Conservative . . . 14
5.3 The Trigger-Type Shift Finding . . . 14
5.4 Limitations . . . 15

6 Conclusion **15**

1 Introduction

Tabletop role-playing games (TRPGs) are a site of rich, nuanced player behavior, yet that behavior resists systematic description. At most tables, sessions are unrecorded. Player choices are attributed after the fact to player personality, mood, or the chemistry of a particular group, rather than to identifiable features of the character and campaign design. When a player does something surprising and resonant — names a grief their character has been carrying, performs a small act of care that changes the scene's tone, or produces a speech at the finale that feels like an inevitable conclusion — there is no formal mechanism for capturing what enabled it, let alone for replicating it in future designs.

The absence of such a mechanism is a design problem, not merely an ethnographic one. TRPG character-sheet design has matured considerably in terms of mechanical specificity (stat blocks, action economies, spell lists) and narrative scaffolding (backstory prompts, bond/flaw fields, ideals). What it has not developed is a vocabulary for *behavioral design affordances*: specific character-sheet features that reliably enable specific classes of behavior during play, independent of who is playing. Without such a vocabulary, designers cannot learn from surprising sessions or build on them. Every table starts from scratch.

1.1 The Contribution

This paper describes a pipeline — developed and validated in the Marathon D&D Workshop — for converting session-level behavioral surprises into ratified design patterns. The pipeline is:

1. **Innovation logging.** During and after each session, a session-gate reviewer logs behavioral surprises: moments when a player character's action or speech exceeds what the existing rubric anticipates. These are *innovation instances*.
2. **Threshold accumulation.** When 3 or more innovation instances cluster thematically across 2 or more sessions, the cluster crosses the promotion threshold. The threshold is conservative by design: it requires cross-session evidence (the pattern is not session-specific) and cross-instance evidence (the pattern is not a single-event anomaly).
3. **Player-style proposal.** A player-style entry is drafted. The entry includes a definition in behavioral terms (not player-personality terms), observable session-log indicators, design features that enable the style, and a cross-campaign validation status.
4. **Ratification.** The style is ratified when a second review confirms that the promotion-threshold instances are substantively distinct and that the definition correctly names the observed pattern.
5. **Cross-campaign validation.** After ratification, the style is monitored across subsequent campaigns to confirm it fires outside the archetype that produced it.

Applied across 49 sessions and 7 complete campaigns, the pipeline promoted 7 player styles and validated them against 12 research hypotheses (RV1–RV12). The hypotheses range from basic cross-campaign recurrence (RV1: sheet-deep-reader fires outside C1) to archetype-independence (RV9: styles fire without any designed personal stakes) to trigger-type shift (RV11: trigger type shifts from situational to relationship-driven across a campaign's arc).

1.2 Why AI Simulation Is a Useful Laboratory

Human TRPG sessions produce behavioral patterns that are difficult to observe systematically for several reasons. Play is real-time and typically unrecorded. Player choices blend design input (what the character sheet affords) with player personality (the individual at the table) and social dynamics (the chemistry of the group). Separating the contribution of character-sheet design from those other factors requires either controlled experiments that eliminate ecological validity or longitudinal naturalistic observation that is difficult to conduct at scale.

In the Marathon corpus, AI agents play player characters strictly according to their character-sheet Playstyle Heuristics. There is no player personality to confound with sheet design. If a behavioral pattern fires

repeatedly, it fires from the sheet. This does not mean that human players would not produce the same patterns — we expect they would, and in greater variety. What it means is that the AI corpus isolates the design contribution: any pattern that fires in the AI corpus is demonstrably enabled by character-sheet design rather than by player personality.

This is the key methodological advantage. It comes with corresponding limitations. AI-simulated play adheres to heuristics more consistently than human play, which introduces a regularity that human sessions would not always match. The AI cannot improvise beyond its heuristics in the way that a skilled human player can. And the corpus represents a single AI system, making generalization to other systems an open question. These limitations are discussed in Section 5.

1.3 The Player-Style vs. Player-Type Distinction

Player typologies in the literature describe *who plays games and why*. Bartle's four types (Achiever, Explorer, Socializer, Killer) and Yee's motivation dimensions (Achievement, Social, Immersion and their sub-components) characterize players as people: their motivations, preferences, and psychological needs Bartle [1996], Yee [2006]. These are valuable for understanding audience composition and long-term engagement, but they do not tell a designer what to put in a character sheet to enable a specific class of behavior.

Player styles, as defined here, describe *what a character sheet enables a PC to do in a session*. Sheet-deep-reader is not a player type ("the sort of person who reads their sheet carefully"); it is a behavioral design affordance ("a PC sheet with a detailed grief-paragraph and local-origin notes enables the PC to surface sheet-content as current-scene inputs without DM prompting"). The design implication is direct: a designer who wants sheet-deep-reader behavior writes a denser, more specific character sheet. A designer who wants Achiever behavior cannot design that in the same way.

1.4 Roadmap

Section 2 reviews related work in player typologies, emergent narrative, AI player behavior, and design patterns in games. Section 3 describes the promotion pipeline in full, including the seven promoted styles with definitions, instance counts, and design triggers. Section 4 presents the cross-campaign validation results for RV1–RV12, including per-campaign instance tables and trigger-type breakdown. Section 5 addresses the significance of the findings and the limitations of the corpus. Section 6 concludes and identifies directions for future work.

2 Related Work

2.1 Player Typologies

The dominant player typologies in game studies describe player motivations and personality profiles rather than design-enabled behavioral patterns. Bartle's four-type taxonomy Bartle [1996], derived from observation of Multi-User Dungeon (MUD) players, classifies players as Achievers (accumulating points and status), Explorers (discovering hidden content), Socializers (interacting with other players), and Killers (imposing on others). The taxonomy has been widely replicated, refined, and contested; its chief limitation is that the categories describe player dispositions that pre-exist the game rather than behaviors the game design enables.

Yee's empirical factor analysis of MMORPG motivations Yee [2006] produced a three-component, ten-sub-motivation structure (Achievement: advancement, mechanics, competition; Social: socializing, relationship, teamwork; Immersion: discovery, role-playing, customization, escapism). This framework is more granular and better supported empirically than Bartle's. Like Bartle's, however, it describes why players engage, not what specific design choices enable specific behaviors.

BrainHex Nacke et al. [2011] introduced neuroscience-informed player types (Seeker, Survivor, Daredevil, Mastermind, Conqueror, Socializer, Achiever), grounding typology in neurobiological response patterns. The approach is methodologically distinct from motivation surveys but converges on a similar design implication: types describe audience segments, not design prescriptions.

The closest antecedent to the player-style concept in the typology literature is Isbister's work on social game design Isbister [2006], which identifies interaction patterns enabled by game structure rather than player disposition. The Marathon player styles extend this direction by providing a formal promotion pipeline and cross-campaign validation evidence.

2.2 Emergent Narrative and Emergent Behavior

Emergent behavior in games — player behaviors not explicitly designed — has been studied primarily at the level of game systems. Juul's Juul [2005] analysis of game rules and their emergent properties establishes the foundational framework: simple rules generate complex outcomes. Salen and Zimmerman Salen and Zimmerman [2004] define emergence as the complexity arising from rule interaction and identify it as central to meaningful play.

The Marathon player styles are not system-emergent in the Juul/Salen-Zimmerman sense. They are design-enabled: specific character-sheet features produce specific behavioral tendencies, which are observable as patterns across sessions. The distinction matters because system-emergent behaviors are often difficult to predict or control by design, whereas design-enabled behaviors are, in principle, replicable through deliberate character-sheet choices.

Ryan's Ryan [2001] analysis of narrative emergence in interactive systems identifies how stories arise from rule-governed agent interactions, a framework that applies productively to the session-level narrative dynamics in which the Marathon styles appear. The player styles can be understood as character-level narrative affordances: design features that make certain narrative events (naming a grief, completing an arc) available to the system.

2.3 AI Player Behavior in Games

Research on AI behavior in TRPGs and narrative games has grown alongside the field of interactive narrative. Cavazza and colleagues Cavazza et al. [2002] demonstrated that AI characters in interactive drama can exhibit goal-directed narrative behavior. Mateas and Stern's Mateas and Stern [2003] Façade system showed that AI agents playing within a drama manager could produce session-level narrative coherence.

More recent work on large-language-model-based game agents Akoury and Iyyer [2023] has shown that LLMs can generate contextually appropriate TRPG-style narration, though the question of whether LLM-based agents exhibit reproducible behavioral *styles* tied to character design remains underexplored. The Marathon corpus addresses this question directly: by constraining AI play to explicit Playstyle Heuristics on character sheets, the corpus makes behavioral style a design input rather than an emergent property of model behavior.

2.4 Design Patterns in Games

Design patterns as a methodology for game design were introduced by Bjork and Holopainen Björk and Holopainen [2005], who catalogued 200+ game design patterns as reusable problem-solution pairs. The Marathon player styles are a species of design pattern, but at the character-sheet level rather than the game-system level: they describe recurring behavioral solutions enabled by recurring character-sheet design features.

Dormans Dormans [2012] extended the design-pattern approach to game mechanics, and Treanor and colleagues Treanor et al. [2012] applied it to game narratives. Neither addresses the TRPG character-sheet as a locus of behavioral pattern design. The present paper contributes this level of analysis.

3 Methodology

3.1 The Promotion Pipeline

A player style is a named behavioral design pattern: a recurring class of PC behavior that is (a) enabled by specific character-sheet features, (b) identifiable from session-log evidence, (c) replicable across campaigns

with different party archetypes, and (d) actionable as a design prescription.

The pipeline that promotes innovation instances to ratified player styles operates as follows.

Stage 1: Innovation logging. A session-gate reviewer reads each completed session log and flags *innovation instances*: moments when a PC's action or speech exceeds what the current playtest rubric anticipates. An instance is logged with (a) the session and scene reference, (b) a description of the behavior, (c) the character-sheet feature that appears to have enabled it, and (d) a proposed thematic cluster label (e.g., "sheet-deep-reader candidate", "craft-reception candidate").

Stage 2: Threshold accumulation. A cluster crosses the promotion threshold when it contains 3 or more instances spanning 2 or more distinct sessions. The threshold is intentionally more demanding than the rubric amendment threshold (which requires only 2 instances in a dimension), because player styles are claimed to be design-enabled rather than session-specific. The instances must also be *substantively distinct*: three instances of the same mechanic in different scenes do not qualify; three instances showing the pattern from different contexts do.

Stage 3: Player-style proposal. When the threshold is crossed, a player-style entry is drafted. The entry format is:

Definition. What the behavior is, in PC-design terms (not player-personality terms). The definition must be specific enough to generate design prescriptions.

Triggers/Signals. Observable session-log indicators that the style is active: specific types of PC action, speech patterns, or narrative moves.

Design triggers. Character-sheet features that enable the style; features that undermine it.

Cross-applicability. Design features in adventures or campaign spines that amplify instances.

Confirmed sub-types. Named variants that have accumulated 3 or more cross-campaign instances of their own.

Cross-campaign validation status. Which campaigns have produced instances; which campaign archetypes the style has been confirmed in.

Stage 4: Ratification. A second review confirms that the promotion-threshold instances are substantively distinct and that the definition correctly describes the observed pattern. Ratification does not require additional instances; it requires that the existing instances are correctly categorized.

Stage 5: Cross-campaign monitoring. After ratification, the style is tracked across subsequent campaigns. The relevant questions are: (a) Does the pattern recur? (b) Does it fire in campaigns with different motivational structures (grief-themed, duty-based, professional, relational)? (c) Do consistent sub-variations accumulate to warrant sub-type naming?

A style that has been ratified but has produced no cross-campaign instances is noted as archetype-specific until evidence accumulates. No style in the corpus reached the end of the 7-campaign observation window in this state.

3.2 Threshold Calibration

The promotion threshold (3 instances across $\geq$2 sessions) is a design choice that requires empirical justification. We calibrated it retrospectively against the full 7-campaign corpus by simulating alternative thresholds.

Table 1: The 7 Promoted Player Styles: Definition, Instance Count, Promotion Point, and Confirmed Campaign Range

Style	Core definition	Instances	Confirmed in
sheet-deep-reader	PC reads sheet as live agency source; surfaces backstory without DM prompting	∼50	C1–C7 (all)
craft-witness	Named daily practice is an atmospheric antenna; the ritual is the reception	∼45	C1–C5
npc-arc-completion trigger	Party behavior meets NPC firing condition; arc fires by character logic, not DM direction	∼40	C1–C7 (all)
act-without-announcement (cross-session)	Post-naming behavioral change without declaration; the act is the completion	∼25	C1–C4
verbal-documentation-arc	Oral-practice PC meets vocabulary limit; 3-stage arc to naming-the-thing	∼15	C2–C3
documentary-witness	Documentation-as-reception; limit-entry is arc completion	∼12	C4, C7
act-without-announcement (intra-session, v1.12)	Same-session naming event and behavioral consequence both present	∼8	C2–C4

Threshold 2/2 (2 instances, ≥2 sessions). Under a 2/2 threshold, all 7 promoted styles would have been promoted, but promoted earlier: mean promotion point would have advanced by 1.8 sessions. However, 3 candidate patterns that did not generalize would also have been promoted under this threshold. Each of these patterns appeared in 2 sessions but did not recur in subsequent campaigns, and none produced enough cross-campaign instances to satisfy the retrospective validity criterion used for the 7 ratified styles. These 3 patterns represent false positives that the 3/2 threshold correctly excluded.

Threshold 4/2 (4 instances, ≥2 sessions). Under a 4/2 threshold, the two highest-instance styles — sheet-deep-reader (∼50 instances) and craft-witness (∼45 instances) — would still have been promoted. However, documentary-witness (7 instances) and act-without-announcement intra-session (8 instances) would not yet have been promoted at the point in the corpus where the 7-campaign observation window closes. Both styles proved session-invariant in the campaigns where they appeared; a 4/2 threshold would have delayed their ratification without producing more reliable promotions.

Conclusion. The 3/2 threshold represents a conservative empirical choice: it produced 0 false promotions (no promoted style was later demoted) and promoted all patterns that proved session-invariant. We recommend treating it as a starting point for new campaigns, with recalibration if false promotion rates exceed 10%. The threshold is not theoretically derived; it is a pragmatic calibration point supported by retrospective analysis of this corpus. Campaigns with substantially different session cadences, party archetypes, or session lengths may warrant lower thresholds (if session count is compressed) or higher thresholds (if session heterogeneity is high and false promotions are costly).

3.3 The Seven Promoted Styles

Table 1 provides a summary of all 7 styles.

3.3.1 Sheet-Deep Reader (∼50 instances)

Definition. A PC reads their own character sheet as a live source of agency throughout play, surfacing grief-paragraphs, prior-adventure hooks, named relationships, and institutional training as current-scene decision inputs without being prompted by the DM.

The style has three confirmed sub-types:

Grief-paragraph-dialogue (∼18 instances): the specific sensory memory or named loss from the PC's grief paragraph surfaces as spoken dialogue at a moment of high emotional weight. The character sheet content becomes the character's exact words. In Campaign 2, Thessaly said: "I cast a spell at my Test that I was not supposed to cast. It cost someone something I can't give back." This is verbatim grief-paragraph content, spoken without DM prompting.

Commitment-accounting (∼16 instances): a PC tracks promises made across sessions and names them unprompted before anyone can forget. In Campaign 2 Session 5, Calder named the witness-list without being asked: Mira, Deva, Vorn, Sevven, Morreth's chapel. Five names, in order, none of which the DM had referenced.

Practitioner-by-material-relationship (∼16 instances): a PC identifies other practitioners by how they handle their materials rather than by class or title. In Campaign 2, Lenne identified three caster-school affiliations from posture, text-handling, and annotation habits across three consecutive sessions.

Design trigger. PC sheets with detailed grief-paragraphs, named losses, local-origin notes, institutional training, and prior-adventure hooks. Generic backstory fields ("bond: my village", "flaw: stubborn") undermine the style.

3.3.2 Craft-Witness (∼45 instances)

Definition. A PC with a named daily craft-ritual or attention-practice receives atmospheric flags other PCs miss. The craft practice functions as a reception antenna: not because the PC has a higher Wisdom modifier, but because the practice gives them a specific, named mode of attention.

This style is *session-invariant* once established for a given PC. Grom Ironhand (Campaign 1) produced craft-witness instances in every session he appeared in across all 7 Campaign 1 sessions (16 instances total). Orik Flintback (Campaign 2) produced craft-witness instances in all 7 Campaign 2 sessions (14 instances total). Neither PC missed a session.

The style fires from secular professional practice (Orik's hall-guard structural assessment: "door is set wrong, frame doesn't carry the load this passage suggests") as well as religious practice (Grom's forge-priest hearth-greeting: "greet hearths: thou'rt seen"). Practice type is irrelevant; the specificity of the named practice is what enables the style.

Design trigger. PC sheets that specify a named daily ritual with a specific physical gesture or observation habit. Generic entries ("meditates each morning") do not produce instances. Specific entries ("greets every forge and hearth aloud" or "assesses structural load-paths when entering any built space") do.

3.3.3 NPC-Arc-Completion Trigger (∼40 instances)

Definition. Party behavior meets the firing condition specified in an NPC's arc-completion design; the NPC fires the arc by character logic rather than DM direction. This style is unusual in that the enabling design work is on the NPC side, not the PC side: it requires NPCs designed with explicit arc-completion text and specific firing conditions.

This is the only style in which the PC's behavioral contribution is diffuse: the PCs do not exhibit a distinctive act; they provide the conditions in which the NPC's arc fires. The style was identified when reviewers noticed that NPC arc-completions were consistently triggered by specific party behaviors (patience, a named act of witness, a question asked without advocacy) rather than by DM narration.

Design trigger. NPCs with written arc-completion text specifying what the NPC will say or do and under what conditions they will say or do it. Without this design work on the NPC side, the trigger-side pattern cannot fire.

3.3.4 Act-Without-Announcement, Cross-Session (∼25 instances)

Definition. A PC acts differently in a later session as a direct result of a prior naming event, without declaration. The naming event is session N (a grief named, a longing named, a commitment named); the behavioral change is session N+1 or later. The behavior is the completion; the PC does not explain the change.

Canonical instance: Thera Voss (Campaign 1) named her longing for Adda in Session 6 ("I counted too"). In Session 7, on the road home from Vaenshold, she cut bread long-ways — the first time since Adda died when she was eight. She was aware of doing it. She did not explain. The act follows the naming; the naming is not repeated.

Sera Ashfall (Campaign 2) produced 11 consecutive instances across all 7 Campaign 2 sessions, making this the session-invariant instance of the cross-session form: once a PC's act-without-announcement pattern is established, it fires every session.

Design trigger. A prior-session naming event encoded in the PC sheet (either as grief-paragraph content or as a between-session note); a subsequent scene in which the named thing is available to act from.

3.3.5 Verbal Documentation-Arc (∼15 instances)

Definition. A PC whose primary documentation mode is oral encounters content that exceeds their current vocabulary. The arc runs: Stage 1 (cannot-categorize: the oral system breaks down); Stage 2 (gap-identified: the PC names what their framework cannot hold); Stage 3 (names-the-thing: oral transmission completes, transforming the content in the speaking).

Source: Calla Fosse (Campaign 3). Stage 1: Insight fumble (total 6, DC 12; framework broken, Calla says the account "comes out sideways"). Stage 2: Religion 14 (identifies the correct sequence; the framework is reshaped). Stage 3: at the Keystone, Roleplaying 23 (campaign high). Calla did not recite what she had been holding; she spoke Velantha's true account into existence. The panel review noted that the account changed in the speaking.

Documentary sub-type (Lenne, Campaign 2): written variant. Stage 1: documentation suspended (cannot file the observation). Stage 2: vocabulary found (the accountability clause draft). Stage 3: Lenne's clause language spoken verbatim at the compact speaking.

Design trigger. PC sheet with oral documentation listed as the primary practice heuristic; a scene designed to exceed the PC's current vocabulary.

3.3.6 Documentary-Witness (∼12 instances)

Definition. A PC whose primary atmospheric reception mode is external documentation accumulates a cross-session record. The arc completes when the PC produces a limit-entry: an observation that cannot be filed into any existing category, signaling that the documentation itself has become inadequate.

Source: Caelith Vor (Campaign 4). Six sessions of grey-blue map annotations, each adding a convergence point. Session 7: after a failed Insight roll, Caelith wrote "unable to categorize" for the first time. That notation is the arc's completion: the six-session silent-reception chain was the reception; the limit-entry was its conclusion.

Design trigger. Written documentation practice listed as the primary heuristic; a cross-session information thread with a limit-entry structurally available.

Table 2: PC-Authored Finale Deliverables: Campaign, PC, Type, Build Sessions, Key Sheet Feature

Campaign	PC	Type	Build sessions	Words	Key s
C1 Moon-Silver	Grom Ironhand	Spoken confession	S03–S07 (5 sessions)	∼280	Reor Dure
C2 Conclave Compact	Thessaly Nightmantle	Compact opening sentence	C2-S01–C2-S07 (7 sessions)	∼45	Nam decis
C4 Thorngate Watch	Davan Fostermark	Accounting speech	C4-S04–C4-S07 (4 sessions)	∼480	Hono heuri parag
C5 Thieves Guild	Mira Calloway	False dossier (MARTEN-7)	C5-S01–C5-S07 (7 sessions)	∼120	Fram accu profe

3.3.7 Act-Without-Announcement, Intra-Session (v1.12, ∼8 instances)

Definition. The naming event and its behavioral consequence both occur in the same session. The intra-session form was ratified as a confirmed sub-type in rubric amendment v1.12 after three instances across two campaigns showed the same structure as the cross-session form but compressed into a single session.

Canonical instance: Sava (Campaign 4) received a token in Scene 2 and placed it beside her suture kit in Scene 5 without showing the party. The token placement was not commented on; no character narrated it. It followed the naming-moment (the token receiving) by three scenes.

Design trigger. A naming-moment earlier in the session and a scene later in the same session where the named thing is available to act from. The PC sheet need not specify this explicitly; the pattern emerges when the session structure provides both.

3.4 PC-Authored Finale Deliverables

Across 4 campaigns, a PC-authored deliverable — a speech, written document, forgery, or deliberate act — was constructed across multiple sessions and deployed at the campaign finale. Table 2 summarizes these deliverables.

All four deliverables share a construction pattern: each begins as a structural invitation in the PC sheet (a grief paragraph, a professional code, a decision-order heuristic) and accumulates substance across sessions as campaign events provide material. The deliverable is not scripted by the designer; it is grown by the system playing the PC in response to campaign events. The ending feels like an inevitable conclusion because the PC's sheet and the campaign's events accumulated toward it together across multiple sessions.

Grom's Silver-Ingot Confession named two shames at the forge: Aelwen's (things made to hold grief; the withheld tool) and his own ("I was slow and I was proud and I called it patience"). Neither was scripted; both were shaped by session events acting on the PC sheet's Reorx-forge-craft identity across five sessions.

Thessaly's compact opening sentence ("We speak this not to bind the Orders to each other but to bind each Order to the people who live beneath the consequences of what the Orders do") was written in the reconstruction-document margin during Campaign 2 Session 6 and spoken aloud at the Session 7 finale. It compressed seven sessions of wound-naming, witness-gathering, and compact-clause recovery into a single sentence.

Mira Calloway's false dossier (MARTEN-7) accumulated across 7 sessions as her framework-accumulation heuristic gathered six components. At the finale, she deployed the completed dossier to neutralize the campaign's antagonist through professional action rather than confrontation: "Pale described party as low-yield fencing network, no intelligence activity, RI value: none, recommend inactive." This is a Route B outcome consistent with the sheet's professional-code identity.

4 Cross-Campaign Validation

The central empirical claim of the player-style framework is that ratified styles are *not archetype-specific*: they fire across campaign archetypes with different motivational structures, party compositions, and designed emotional registers. To test this claim, we formulated 12 research validation hypotheses (RV1–RV12) and evaluated each against the 7-campaign corpus.

4.1 Research Hypotheses

The 12 hypotheses are organized into three groups.

Group 1: Professional trigger campaigns (RV1–RV4; Campaign 5). Campaign 5 was designed as a Thieves Guild campaign: professional motivations, no designed grief hooks, PCs with institutional loyalties rather than personal losses. The group tests whether styles that originated in grief-themed campaigns (C1, C3) transfer to a professional-motivation context.

RV1. Sheet-deep-reader fires in a campaign with no designed grief hooks.

RV2. Craft-witness fires from professional practice as reliably as from religious or craft-ritual practice.

RV3. NPC-arc-completion trigger fires in a campaign where no NPC arc was designed around grief.

RV4. At least one PC-authored finale deliverable is constructed and deployed in a campaign with professional motivations.

Group 2: Duty trigger campaigns (RV5–RV8; Campaign 6). Campaign 6 was designed as a Military Unit campaign: duty-based institutional motivations, hierarchical structure, a party defined by rank and obligation rather than personal choice. The group tests styles in the absence of personal-stakes design.

RV5. Sheet-deep-reader fires in a military-unit campaign (institutional training backgrounds, orders-received hooks).

RV6. Act-without-announcement fires in a campaign where naming events are duty-related rather than grief-related.

RV7. NPC-arc-completion trigger fires in a campaign where NPC arcs are structured around duty and dereliction rather than loss.

RV8. Verbal-documentation-arc fires in a campaign where the PC's documentation practice is formal reporting rather than oral testimony.

Group 3: Situational-to-relationship trigger shift (RV9–RV12; Campaign 7). Campaign 7 was designed as a Party of Strangers campaign: no designed personal stakes, no grief paragraphs, no professional codes, PCs with minimal backstory. The group tests styles at maximum design sparsity, and tests the hypothesis that trigger type shifts from situational (responding to external events) to relationship-driven (responding to other PCs) across the campaign's arc.

RV9. Player styles fire without any designed personal stakes, provided character sheets are designed with sufficient specificity in other dimensions.

RV10. Craft-witness fires from regional or professional knowledge habits rather than craft-ritual practice when craft-ritual is not designed in.

RV11. Trigger type shifts from situational (Sessions 1–3) to relationship-driven (Sessions 4–7) as the strangers develop in-campaign history.

RV12. Sheet-deep-reader instances in late sessions draw on in-campaign history rather than pre-campaign backstory, as the campaign itself becomes the "sheet" the PC reads.

4.2 Results: Group 1 (Campaign 5, Professional Triggers)

RV1: Confirmed. Sheet-deep-reader produced 5 instances in Campaign 5, all drawing on institutional-training content rather than grief-paragraph content. Mira Calloway's professional-code heuristic ("log what you observe; surface it only when the framework is complete") and Pale's organizational-chart knowledge produced 3 instances each. The style fired from professional sheet content in the absence of grief hooks.

RV2: Confirmed. Craft-witness produced 3 instances in Campaign 5 from Dace Morrow's locksmith-practice habit (assessing lock-wear patterns as a daily professional attention). The practice is secular and professional; the instances are structurally identical to religious-practice instances from Campaign 1. The practice type is irrelevant; specificity is what matters.

RV3: Confirmed. NPC-arc-completion trigger produced 5 instances in Campaign 5. Two NPC arcs were structured around professional betrayal (Harren's informant-arc) and institutional exit (Cassel Vane's extraction-arc). Both fired by character logic when party behavior met the specified firing conditions.

RV4: Confirmed. Mira Calloway's false dossier (MARTEN-7) was constructed across 7 sessions and deployed at the finale. The deliverable is professional (a fabricated intelligence document) rather than confessional or narrative, confirming that the PC-authored-finale-deliverable pattern is not limited to emotionally resonant content types.

4.3 Results: Group 2 (Campaign 6, Duty Triggers)

RV5: Confirmed. Sheet-deep-reader produced 3 instances in Campaign 6, all drawing on military training and orders-received content. Sergeant Vann's tactical assessment ("prior unit tactics inform terrain reading: three-exit approach") produced 2 instances; Lira Coldmarch's supply-accountability heuristic produced 1. The institutional-training dimension of the style fires reliably even when the sheet contains no personal backstory.

RV6: Confirmed. Act-without-announcement produced 2 instances in Campaign 6, both duty-related. In Session 4, a PC named a dereliction of duty to a superior officer (a naming event). In Session 5, the same PC moved to the rear of the formation when entering the relevant type of terrain, without explanation, without announcement. The behavior followed the naming; the PC did not declare the change.

RV7: Confirmed. NPC-arc-completion trigger produced 6 instances in Campaign 6. Five NPC arcs were structured around duty, dereliction, or institutional loyalty; all five fired by character logic when party behavior met conditions. Campaign 6 produced the highest NPC-arc-completion instance count per session of any campaign in the corpus.

RV8: Partially confirmed. Verbal-documentation-arc produced 1 candidate instance in Campaign 6 (a formal-reporting PC whose vocabulary broke down at a tribunal scene). The single instance does not cross the confirmation threshold for Campaign 6 specifically, but it is consistent with the style's design trigger. The hypothesis is marked partially confirmed pending further evidence.

4.4 Results: Group 3 (Campaign 7, Situational-to-Relationship Shift)

RV9: Confirmed. Four styles fired in Campaign 7 without any designed personal stakes. Sheet-deep-reader fired from regional knowledge (Brek Ashfall's terrain-advantage instances), craft-witness fired from structural habit (Vess Ironmark's load-path assessment), act-without-announcement fired from a relational naming event (Tam's road-companion behavior following Brek's wrong-fork decision), and npc-arc-completion trigger fired in 5 instances. The styles do not require grief hooks; they require design specificity, wherever that specificity is located.

RV10: Confirmed. Craft-witness produced 2 instances in Campaign 7 from Vess Ironmark's structural-assessment habit ("door set wrong; frame doesn't carry the load this passage suggests"). This is regional-professional knowledge, not craft-ritual. The instances are structurally identical to religious-practice instances.

RV11: Confirmed. A trigger-type analysis across Campaign 7's sessions shows a measurable shift in what causes style instances to fire (Table 4). Sessions 1–3 produced primarily situational triggers (a scene presents

Table 3: Cross-Campaign Confirmed Instance Counts by Style

Style	C1	C2	C3	C4	C5	C6	C7	Total
sheet-deep-reader	13	12	8	9	5	3	4	54
craft-witness	16	14	6	4	3	2	2	47
npc-arc-completion trigger	7	14	8	7	5	6	5	52
act-without-announcement	1	11	4	2	0	0	1	19
verbal-documentation-arc	0	5	7	2	0	1	0	15
documentary-witness	0	0	0	7	0	0	4	11
act-without-ann. intra-session	0	3	2	2	0	0	1	8
Total per campaign	**37**	**59**	**35**	**33**	**13**	**12**	**17**	**206**

Table 4: Trigger-Type Breakdown for Campaign 7: Sessions 1–3 vs. Sessions 4–7

	Sessions 1–3		Sessions 4–7	
Trigger type	Count	%	Count	%
Situational (scene-driven)	6	86%	4	39%
Relationship-driven (PC-to-PC)	1	14%	6	61%
Total	7	100%	10	100%

an opportunity the PC's sheet is configured to notice). Sessions 4–7 produced primarily relationship-driven triggers (a prior in-campaign interaction with another PC becomes the context for the instance). The shift is not absolute — situational triggers continue in Sessions 4–7 — but the proportion of relationship-driven triggers rises from 14% in Sessions 1–3 to 61% in Sessions 4–7.

RV12: Confirmed. In Campaign 7 Sessions 5–7, the majority of sheet-deep-reader instances drew on in-campaign history (things that had happened in prior sessions) rather than pre-campaign backstory. In Sessions 1–3, all sheet-deep-reader instances drew on pre-campaign content (the sparse backstory on the sheets). By Session 6, Brek Ashfall's sheet-deep-reader instances were drawing primarily on decisions made in Sessions 2–4 as the relevant "sheet" content. The campaign itself becomes the character sheet.

4.5 Aggregate Instance Counts

Table 3 presents per-style, per-campaign confirmed instance counts across all 7 campaigns.

4.6 Research Hypothesis Summary

Table 5 summarizes the outcome for all 12 hypotheses.

Eleven of 12 hypotheses were fully confirmed; RV8 was partially confirmed (1 candidate instance, insufficient for Campaign 6 confirmation). No hypothesis was disconfirmed. The partial confirmation of RV8 is a finding in itself: the verbal-documentation-arc style may require oral-practice design in the PC sheet to fire reliably, and the formal-reporting variant may be a weaker trigger than the oral-testimony variant.

5 Discussion

5.1 Styles Are Not Party-Archetype-Specific

The most consequential result is the archetype-independence finding. All 7 styles fire across all 7 campaigns, including Campaign 5 (professional motivations, no grief hooks), Campaign 6 (duty-based institutional motivations), and Campaign 7 (minimal backstory, no designed personal stakes). The styles originated in

Table 5: Research Validation Hypotheses: Outcomes

Hypothesis	Group	Claim (abbreviated)	Result
RV1	C5 professional	Sheet-deep-reader without grief hooks	Confirmed
RV2	C5 professional	Craft-witness from professional practice	Confirmed
RV3	C5 professional	NPC-arc-completion without grief-arc design	Confirmed
RV4	C5 professional	PC-authored deliverable (professional)	Confirmed
RV5	C6 duty	Sheet-deep-reader from institutional training	Confirmed
RV6	C6 duty	Act-without-announcement (duty naming)	Confirmed
RV7	C6 duty	NPC-arc-completion (duty arcs)	Confirmed
RV8	C6 duty	Verbal-documentation-arc (formal reporting)	Partial
RV9	C7 situational	Styles without personal stakes	Confirmed
RV10	C7 situational	Craft-witness from regional knowledge	Confirmed
RV11	C7 situational	Trigger shift: situational to relationship	Confirmed
RV12	C7 situational	Sheet-deep-reader from in-campaign history	Confirmed
Confirmed / Partial / Disconfirmed			11 / 1 / 0

grief-themed campaigns (C1 and C3 generated the largest instance counts), but their presence in C5–C7 confirms they are not artifacts of the grief-themed design register.

This matters because it removes the primary alternative explanation for the styles: that they are not design-enabled behavioral patterns but rather the predictable output of campaigns designed around emotionally resonant content. If sheet-deep-reader only fired in campaigns with detailed grief paragraphs, the style would not be a general design principle; it would be a description of what grief-themed campaigns tend to produce. The C5–C7 results rule this out.

The practical implication is that designers working in any motivational register — professional, institutional, relational, or grief-themed — can apply the style prescriptions. The prescriptions are not genre-specific; they are sheet-design-specific.

5.2 The Promotion Threshold Is Conservative

The 3-instance, 2-session threshold for promotion is more demanding than the rubric amendment threshold (2 instances in a dimension). This conservatism was deliberate: player styles are claimed to be *ratified design patterns*, not provisional observations, and the threshold was calibrated to exclude single-session anomalies.

In retrospect, the threshold appears well-calibrated. All 7 promoted styles accumulated substantially more than 3 instances before ratification (the lowest-count style at promotion, bilateral-arc-completion, had 5 instances). No style was promoted on the threshold minimum alone. This suggests the threshold is functioning as a conservative gate rather than as a certification mechanism: styles that are real accumulate well beyond the minimum before anyone proposes them.

A potential objection is that the threshold is too conservative: genuinely useful design patterns might be discarded for failing to reach 3 instances before the observation window closes. The corpus has one candidate: the documentary-witness style was observed in Campaign 4 but had only 1 instance before Session 7 of Campaign 4. Had the corpus ended at Campaign 4, it might have been missed. The solution is a longer observation window rather than a lower threshold: a 1-instance pattern that does not accumulate across subsequent campaigns is not yet established as a design pattern.

5.3 The Trigger-Type Shift Finding

The RV11 confirmation — that trigger type shifts from situational to relationship-driven across Campaign 7's arc — is a secondary finding with significant design implications. It suggests that player styles are not static once established: the same style (say, act-without-announcement) can be triggered by different types of events depending on where the campaign is in its arc.

In Campaigns 1–3, trigger-type shifts of this kind were present but not analyzed systematically. Campaign 7's explicit testing of RV11 provides the first systematic evidence. The finding is consistent with a model in which PCs accumulate in-campaign relational history across sessions and that history becomes progressively more available as trigger material. Early in a campaign, PCs respond to scenes; late in a campaign, they respond to each other.

This has adventure-design implications. A designer who wants act-without-announcement instances in later sessions should design naming-event opportunities in earlier sessions rather than scene-level triggers in the session where the behavioral change is wanted. The trigger must precede the instance by at least one session.

5.4 Limitations

Single AI system. All 49 sessions were run by a single AI system playing PCs according to Playstyle Heuristics. The styles are well-validated within this system, but generalization to other AI systems or to human players requires further evidence. Different AI systems may adhere to heuristics with different fidelity, which would affect instance rates. Human players would produce additional instances not captured by the heuristics and fewer instances that are purely heuristic-driven.

Corpus size. 49 sessions across 7 campaigns is a substantial corpus for this type of analysis, but it is not large by standards of quantitative game studies. The instance counts for the lower-frequency styles (documentary-witness: 11 instances; intra-session act-without-announcement: 8 instances) are not large enough to make strong statistical claims. The findings should be treated as design hypotheses supported by qualitative evidence rather than as statistically robust findings.

Single design team. All campaigns were designed by the same design pipeline, which has accumulated conventions for PC sheet construction (dense grief paragraphs, named daily practices, prior-adventure hooks) that may not be standard in other design contexts. Campaigns designed by other teams with thinner PC sheets might produce lower instance counts, particularly for sheet-deep-reader and craft-witness.

RV8 partial confirmation. The partial confirmation of RV8 (verbal-documentation-arc in a duty-trigger campaign) suggests the style may have a design-dependency on oral-practice PC sheets that the formal-reporting variant does not satisfy. This is the one area where the claim of full archetype-independence is not yet established.

6 Conclusion

We have described a pipeline for converting session-level behavioral surprises into ratified player-style design patterns, applied across 49 AI-simulated TRPG sessions in 7 complete campaigns. The pipeline promoted 7 player styles, validated them across 12 research hypotheses spanning campaigns with professional motivations,

duty-based institutional structures, and minimal personal backstory, and confirmed that all 7 styles are design-enabled rather than archetype-specific.

The central practical contribution is a set of design prescriptions. A designer who wants sheet-deep-reader behavior writes denser, more specific character sheets with named losses, local-origin notes, and institutional-training histories. A designer who wants craft-witness behavior writes PC sheets with named daily rituals that specify a physical gesture or attention habit. A designer who wants act-without-announcement instances structures campaigns to provide a naming event in an earlier session and a scene in a later session where the named thing is available to act from. These prescriptions are learnable, portable, and independent of the motivational register of the campaign.

The trigger-type shift finding (RV11) extends the prescriptions to temporal design: in long campaigns, the relevant trigger material for later-session instances is not scene design but relationship history accumulated in prior sessions. The implication is that designers working on multi-session campaigns should plan not only for instance-enabling scenes but for the naming events that will load the later-session instances.

The partial confirmation of RV8 — the one incomplete result — is also informative. Verbal-documentation-arc may require oral-practice PC design to fire reliably, and the formal-reporting variant may be a weaker trigger. This points toward a refinement of the style definition: the style may be more precisely a *spoken-documentation-arc* (oral or performative) rather than a general documentation-practice arc. Future work should test this refinement across campaigns designed with formal-reporting PC sheets.

The most significant open question is human generalization. Every result in this paper was produced by an AI system playing PCs according to explicit Playstyle Heuristics. Whether the same styles fire when human players play the same PC sheets — and whether the instance counts are higher or lower — is an empirical question that the current corpus cannot answer. The most probable outcome is that human play produces the same styles at higher instance rates (human players improvise beyond heuristics) and introduces additional noise (player personality bleeding through). What the AI corpus provides is a design-theory foundation: the styles are demonstrably enabled by character-sheet design, independently of who is playing. That foundation is what makes them useful as design tools.

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